

Delete file

Exception 1 → FileNotFoundError

Exception 2 → PermissionError

Numpy in Python

Numerical Python → Scientific

Computing in Python

ndarray

Different datatype

Same datatype

Array

Lists

Homogeneous type

Heterogeneous type

1 element

≠ elements

1D

3D

2D

$\overset{0}{1} \overset{1}{2} \overset{2}{3} \overset{3}{4} \overset{4}{5}$
 [1 2 3 4 5] $\longrightarrow$ (5,) $\longrightarrow$ 1D array

$\longrightarrow$ (3, 3) $\longrightarrow$ 2D array

rows
 # columns

array([[[1, 2],
 [3, 4]],
 [[5, 6],
 [7, 8]]]) $\longrightarrow$ (2, 2, 2)

sub matrices
 # columns
 # rows

array([[[1, 2, 3],
 [3, 4, 5]],
 [[5, 6, 7],
 [7, 8, 9]],
 [[9, 10, 11],
 [11, 12, 13]]])

0 1 2
 0 5 6 7
 1 7 8 9
 0 9 10 11
 1 11 12 13

$\begin{vmatrix} 4 & 5 \\ 8 & 9 \end{vmatrix}$

$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$

$\begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix}$

$\Rightarrow \begin{pmatrix} 1 \times 4 + 2 \times 6 & 1 \times 5 + 2 \times 7 \\ 3 \times 4 + 4 \times 6 & 3 \times 5 + 4 \times 7 \end{pmatrix}$

array $\longrightarrow$ Element wise operation

$\begin{vmatrix} 1 \times 4 & 2 \times 5 \\ 3 \times 6 & 4 \times 7 \end{vmatrix}$